Lab Task 8: POS Tagging using Viterbi & Log-Linear Model

Aim:

To perform POS tagging using Viterbi Decoding (HMM) and compare it with a LogLinear Model.

Code:

```
from collections import defaultdict, Counter
from sklearn.linear_model import LogisticRegression
from sklearn.feature_extraction import DictVectorizer
from sklearn.pipeline import Pipeline
data = [
    [("The", "DT"), ("market", "NN"), ("is", "VBZ"), ("rising", "VBG")],
    [("She", "PRP"), ("is", "VBZ"), ("happy", "JJ")],
    [("Dogs", "NNS"), ("are", "VBP"), ("running", "VBG")]
]
train = data[:2]
test = data[2:]
emit = defaultdict(Counter)
for sent in train:
    for word, tag in sent:
        emit[tag][word] += 1
def hmm_tag(sentence):
    result = []
    for word in sentence:
        best_tag = max(emit, key=lambda t: emit[t][word])
        result.append((word, best_tag))
    return result
sentence = ["Dogs", "are", "running"]
print("HMM Prediction:")
print(hmm_tag(sentence))
def features(word):
    return {
        "word": word,
        "suffix": word[-2:]
    }
X, y = [], []
for sent in train:
    for word, tag in sent:
        X.append(features(word))
        y.append(tag)

# Model
model = Pipeline([
    ('vec', DictVectorizer()),
    ('clf', LogisticRegression(max_iter=200))
])
model.fit(X, y)
# Prediction
test_words = [w for w, t in test[0]]
pred = model.predict([features(w) for w in test_words])
```

```
print("\nLog-Linear Prediction:")
print(list(zip(test_words, pred)))
```

OUTPUT:

```
HMM Prediction:
[('Dogs', 'DT'), ('are', 'DT'), ('running', 'DT')]
Log-Linear Prediction:
[('Dogs', 'VBZ'), ('are', 'VBZ'), ('running', 'VBG')]
```